demonstrate that one space dimension had been overlooked all along. Thus, Witten showed that the five ten-dimensional frameworks that string theorists had developed for more than a decade were actually five approximate descriptions of a single, underlying eleven-dimensional theory.

You might wonder whether this unexpected realization invalidated previous work in string theory. By and large, it didn't. The newfound tenth spatial dimension added an unanticipated feature to the theory, but if string/M-theory is correct, and should the tenth spatial dimension turn out to be much smaller than all others—as, for a long time, had been unwittingly assumed—previous work would remain valid. However, because the known equations are still unable to nail down the sizes or shapes of extra dimensions, string theorists have expended much effort over the last few years investigating the new possibility of a not-so-small tenth spatial dimension. Among other things, the wide-ranging results of these studies have put the schematic illustration of the unifying power of M-theory, Figure 13.1, on a firm mathematical foundation.

I suspect that the updating from ten to eleven dimensions—regardless of its great importance to the mathematical structure of string/M-theory — doesn't substantially alter your mind's-eye picture of the theory. To all but the cognoscenti, trying to imagine seven curled-up dimensions is an exercise that's pretty much the same as trying to imagine six.

But a second and closely related insight from the second superstring revolution does alter the basic intuitive picture of string theory. The collective insights of a number of researchers—Witten, Duff, Hull, Townsend, and many others—established that *string theory is not just a theory of strings.*

Branes

A natural question, which may have occurred to you in the last chapter, is *Why strings?* Why are one-dimensional ingredients so special? In reconciling quantum mechanics and general relativity, we found it crucial that strings are not dots, that they have nonzero size. But that requirement can be met with two-dimensional ingredients shaped like